

Powering AI: Energy, Control, and Consequence

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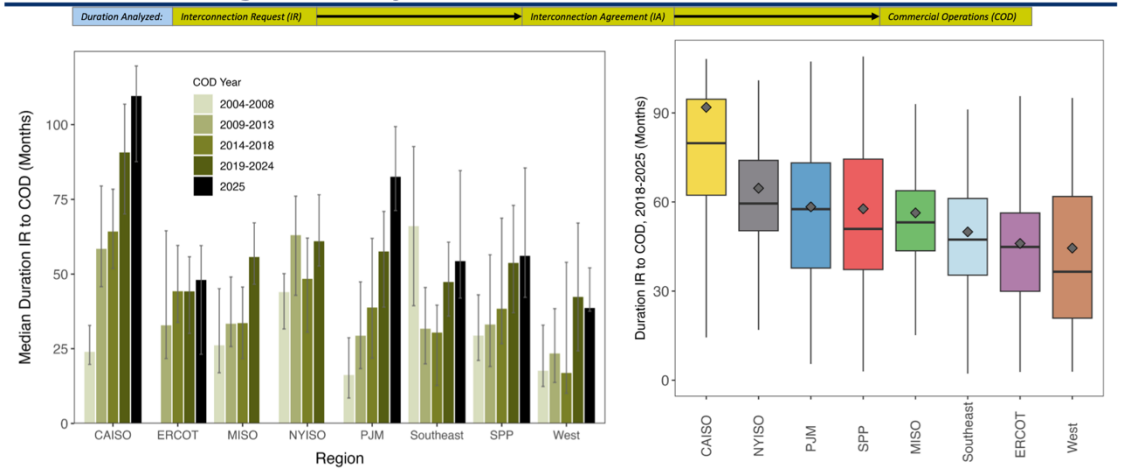
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Behind the Meter Generation

US grid backlog is growing substantially with wait times exceeding 5 years for interconnection

- Transformer shortages
- Public Utility Moratoriums

The request to operational timeline has increased in many regions; median duration since 2018 is longer in FERC-jurisdictional ISOs





Behind the Meter Power generation gained in popularity with data center developers

Large Gas Turbine Manufacturers

GE Vernova, Siemens & Mitsubishi together occupy 75% of the power generation gas turbine market

GE Vernova expects to end 2025 with an 80-GW gas turbine backlog that stretches into 2029

CEO Scott Strazik said he expects gas turbine reservations to be sold out through 2030 by the end of 2026.



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Generation T&D Grid Reliability Electrification Load Management Renewables Storage

DIVE BRIEF

Gas turbine manufacturers expand capacity, but order backlog could prove stubborn

One expert said the industry moves are unlikely to "drastically reduce" wait times, which can stretch as long as seven years.

Published Sept. 5, 2025

OILPRICE.com
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Siemens Energy Reaches Record €136bn Backlog Amid Data Centre Demand Surge

Large gas turbine manufacturers will not be able to meet the demand

Industry is trending towards aeroderivative – aircraft engines adapted for power generation



Gas turbines outside the xAI data center in Memphis.Photographer: George Walker IV/AP Photo

xAI data centers



Aeroderivative - Small gas turbines based on aircraft engines

- Transported in trailers
- Installed and produce power in weeks time frame
- Fuel Flexibility – Liquid and Gaseous fuels

Industry is trending towards aeroderivative – aircraft engines adapted for power generation

CFM56 engines entered service in 1982

Boom Supersonic to Power AI Data Centers with Superpower Natural Gas Turbines; Adds $300 Million in New Funding

Boom secures 1.21 gigawatt order from Crusoe for 29 Superpower units

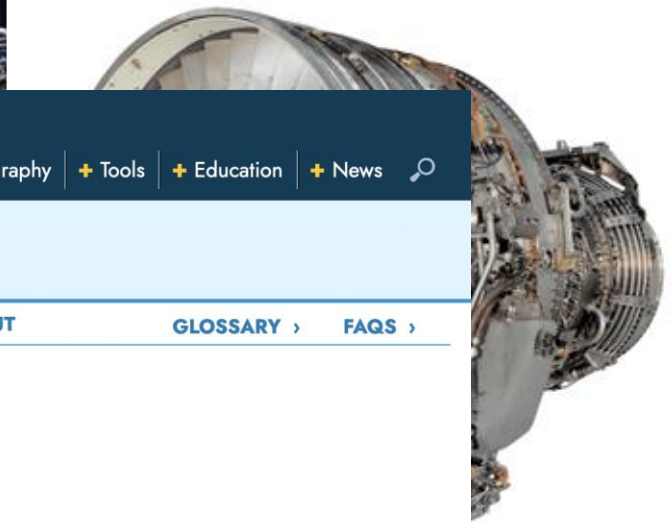


Data Centers Look to Old Airplane Engines for Power › Modified turbofans are helping to power the data-center boom



FTAI unveils CFM56 Power platform

January 2, 2026 11:32 am MRO & Production FTAI Aviation



d to convert CFM56 aircraft engines delivering a flexible, cost-efficient

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IN-BRIEF ANALYSIS

JANUARY 21, 2026

Air power: Tallying electricity generating potential from retired military aircraft

Babcock & Wilcox Select… Generator Sets for Applie…

Posted January 8, 2026

- **Selection will enable B&W**
- **Partnership with Siemens … rapidly growing AI Data Center**

(AKRON, Ohio – January 8, 2026) … has selected Siemens Energy to pro… project to deliver one gigawatt of p… entered into an agreement for a lim… enable B&W to deliver power for th… expected in the first quarter of 202…

B&W previously announced it has s…

Caterpillar Utah data center

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Data Centers

Joule, Caterpillar Partner to Deliver Utah 4-GW Power and Hyperscale Computing Campus

The 4,000-acre project in rural Millard County would be the state's largest data center-power complex

By Bryan Gottlieb



G3520K
Gas Generator Sets

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Key Specs

Continuous Rating
2567 / 2552 ekW @ 1.0pf

Fuel Type
Natural Gas & Propane

Maximum Electrical Efficiency
46.0%

Overview

2552 - 2567 kW Gas Generator

From natural gas to biogas, coal mine gases, or hydrogen fuel blends to support combined heat and power (CHP), emergency power, and local grid support applications, Caterpillar has a wide range of reliable gas power solutions.

Need approximately 1600 generator sets



Powering AI

Aerial view of Microsoft's new AI datacenter campus in Mt Pleasant, Wisconsin, https://blogs.microsoft.com/blog/2025/09/18/inside-the-worlds-most-powerful-ai-datacenter/

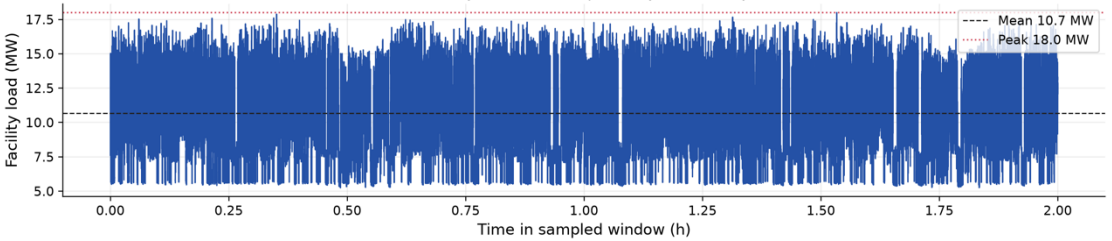
Is total capacity in terms of MW of generation the right metric?

Is it purely a hardware question?

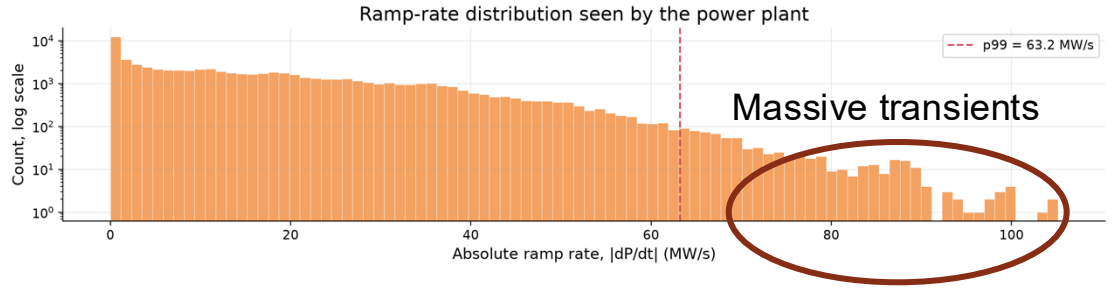
Data Center Load: Islanding Example

MIT Supercloud Load Profile

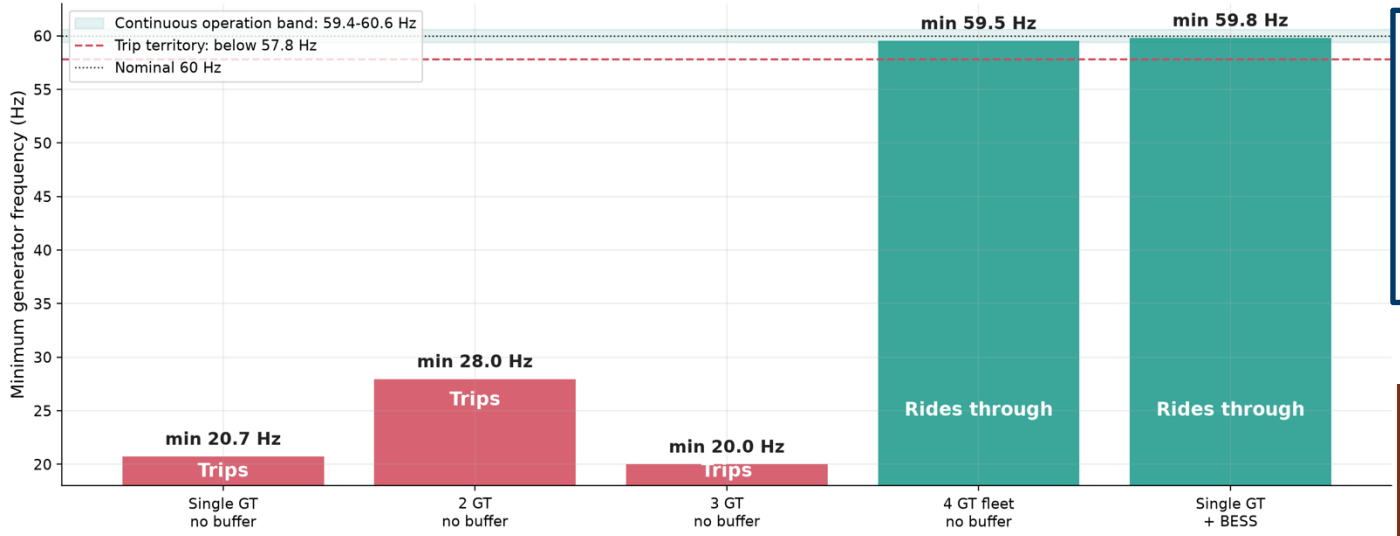
- Per GPU telemetry recorded every 100ms
- Used a 2h, 35 active GPU trace
- Scaled it to match power demand of 18 MW
- Worst case scenario of electrical load from a data center







Data Center Load: Islanding Example



BESS acts as a shock absorber

- 9.9 MW
- 18.4 kWh
- 540 C-rating

Standard lithium-ion 1C - 4C

Need specialized batteries, UPS equipment, flywheels, supercapacitors, or a hybrid

GT = LM2500 aeroderivative, Rated power = 22 MW
BESS = Battery Energy Storage System

1. Focus more on ramp rates instead of total power generation capacity
2. Data centers different from large industrial facilities where electrical load is tied to physical equipment
3. Data center power generation demand can be controlled by software

Ramp Rates and Job Scheduler

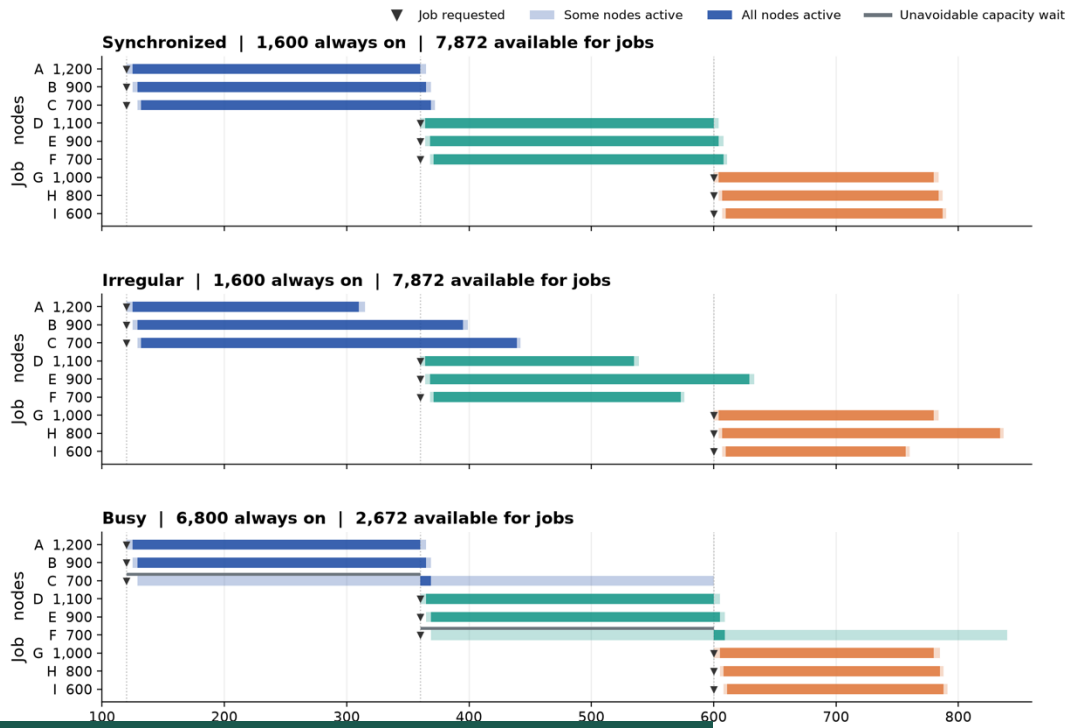
9 batch jobs in 3 waves

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Scheduling Rule	Largest change (MW/s)	Job completion delay (s)
Immediate	6.185	0
512 node wave	1.131	5
256 node wave	0.565	10
226 node wave	0.499	12



Small adjustments in job scheduler can improve the tradeoff between computing service and power system ramp rates

Conclusion

1. U.S. grid-interconnection backlogs are accelerating interest in behind-the-meter generation.
2. Long gas-turbine lead times are encouraging data centers to consider a broader mix of power technologies.
3. Gas-based behind-the-meter generation can supply the energy, but fast buffers, coordinated controls, and power-aware workload scheduling needed to manage rapid AI load changes.
4. The best solution is to combine smarter workloads with generation and storage, reducing how much expensive power generation hardware must be built.

How much generation and storage can be avoided by making AI workloads more flexible for reduced ramp rates?

Thank You

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